On the Expected Coprime Density of the Least Common Multiple of Random Integers

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Abstract

Let $X_1, X_2, \ldots$ be independent and identically distributed random variables uniform on $\{2, 3, \ldots, M\}$, and let $L_k = \operatorname{lcm}(X_1, \ldots, X_k)$. We study the expected coprime density

$$\rho_k(M) = \mathbb{E}\left[\frac{\varphi(L_k)}{L_k}\right],$$

where φ is the Euler totient function. We establish an exact finite-sum identity for $\rho_k(M)$, written as a sum over subsets of primes $p \leq M$ and obtained by Möbius inversion on the subset lattice. The identity implies $\rho_k(M) \to P_M := \prod_{p \leq M}(1 - 1/p)$ as $k \to \infty$ with M fixed, with a sharp asymptotic equivalence $\rho_k(M) - P_M = P_M \Sigma_k(M)(1 + o(1))$ at geometric rate, where $\Sigma_k(M) = \sum_{p \leq M} q_p(M)^k/(p-1)$ and $q_p(M) = 1 - \lfloor M/p \rfloor/(M-1)$. In the joint regime $k = \lfloor cM \rfloor$, $M \to \infty$ (fixed $c > 0$), we prove the full asymptotic equivalence

$$\ln M \cdot \frac{\rho_{\lfloor cM \rfloor}(M) - P_M}{P_M} \longrightarrow A(c) := \sum_{j \geq 1} e^{-cj} \ln\left(1 + \tfrac{1}{j}\right),$$

strengthening the leading-order statement to an $o(1)$ remainder by decomposing the multi-prime correction according to subset cardinality and applying a geometric-mean inequality for subset densities. Numerical computation to $M = 10^7$ across 15 values of $c \in [0.05, 15]$ illustrates the finite-M behaviour; linear extrapolation in $1/\ln M$ agrees with $A(c)$ to within approximately 0.5% for $c \in [0.1, 5]$ and 1.5% over the full tested range. The regime studied here complements earlier random-LCM models by combining i.i.d. sampling with replacement, the observable $\varphi(L_k)/L_k$, and the fixed-support and diagonal scalings considered below.

Keywords: Random LCM, Euler totient, Mertens product, Meissel–Mertens constant, probabilistic number theory, coprime density, Möbius inversion, Rosser–Schoenfeld bounds.
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1 Introduction

The least common multiple of random integers is a classical object. Cesàro [4] studied gcd-averages for uniform random pairs; the expected LCM asymptotic $n^2\zeta(3)/\zeta(2)$ as $n \to \infty$ follows via the identity $\operatorname{lcm}(a, b) = ab/\gcd(a, b)$. The subject has seen substantial recent development. Cilleruelo, Rué, Šarka, and Zumalacárregui [5] established almost-sure asymptotics for $\log \operatorname{lcm}$ of random subsets of $\{1, \ldots, n\}$; Fernández and Fernández [6] surveyed divisibility properties of random tuples; Hilberdink and Tóth [7] obtained the average value of lcm of k integers; Alsmeyer, Kabluchko, and Marynych [1] proved limit theorems for $\log \operatorname{lcm}$ of random sets; Bostan, Marynych, and Raschel [2] established distributional convergence of $f(L_n(k))/n^{r_k}$ for multiplicative functions f of polynomial growth, where $L_n(k) = \operatorname{lcm}(X_1, \ldots, X_k)$ with X_i uniform on

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$\{1,\ldots,n\}$ and k fixed as $n \to \infty$; Buraczewski, Iksanov, and Marynych [3] established a central limit theorem for $\log \operatorname{lcm}$ of a uniformly sampled m-tuple; Iksanov, Marynych, and Raschel [8] extended to hyperbolic regions; Kabluchko, Marynych, and Raschel [9] to broader domains; and Sanna [13] to q-analogues. A dissection of Mertens' prime product formula along a different axis (number of prime factors of summand integers) was given by Lichtman [10].

The literature includes several distinct sampling regimes: fixed-size random tuples as the support grows, Bernoulli random subsets whose cardinality grows with the support, and uniformly sampled tuples in joint limits. The present paper instead focuses on i.i.d. sampling with replacement from $\{2,\ldots,M\}$, first with fixed support and $k \to \infty$, and then in the diagonal regime $k = \lfloor cM \rfloor$ with $M \to \infty$. The natural object of study in these regimes is not the LCM itself — which almost surely equals $\operatorname{lcm}(1,\ldots,M)$ for large k at fixed M — but the arithmetic function

$$\frac{\varphi(L_k)}{L_k} = \prod_{p \mid L_k}\left(1 - \frac{1}{p}\right),$$

which measures the density of integers coprime to L_k. This quantity takes values in $(0,1]$ and its large-k behaviour is governed by exactly which primes $p \le M$ appear as divisors of L_k. One expects the Mertens product as the limit, and this paper makes that expectation precise with explicit convergence rates. Second-order fluctuations — the variance asymptotic and a central limit theorem for the centred log coprime density — are developed in separate work in preparation.

1.1 Main results

Throughout, fix an integer $M \ge 2$. Let $X_1, X_2, \ldots$ be i.i.d. uniform on $\{2,3,\ldots,M\}$, set $L_k = \operatorname{lcm}(X_1,\ldots,X_k)$, and write $\rho_k(M) = \mathbb{E}[\varphi(L_k)/L_k]$. We exclude 1 because it is inert under the least common multiple. Including it would only add null draws. On $\{1,\ldots,M\}$, for example, $q_p(M)$ would become $1 - \lfloor M/p \rfloor / M$; the underlying prime-appearance mechanism is unchanged. Also let $\mathcal{P}_M = \{p : p \le M\}$, $m_p = \lfloor M/p \rfloor$, and

$$q_p(M) = 1 - \frac{m_p}{M-1},$$

the probability that a single uniform draw from $\{2,\ldots,M\}$ is not a multiple of p. For $T \subseteq \mathcal{P}_M$, let

$$\beta_T(M) = \frac{1}{M-1}\left|\left\{n \in \{2,\ldots,M\} : \gcd\left(n, \textstyle\prod_{p\in T} p\right) = 1\right\}\right|.$$

The Mertens product up to M is $P_M := \prod_{p \le M}(1 - 1/p)$.

Theorem 1 (Exact identity). *For every $M \ge 2$ and every $k \ge 1$,*

$$\rho_k(M) = P_M \sum_{T \subseteq \mathcal{P}_M} \beta_T(M)^k \prod_{p \in T} \frac{1}{p-1}. \tag{1}$$

Theorem 2 (Limit and rate). *Let $\Sigma_k(M) := \sum_{p \le M} q_p(M)^k/(p-1)$ and $\Lambda(M) := \sum_{p \le M} 1/(p-1)$. For every fixed $M \ge 2$, $\lim_{k\to\infty} \rho_k(M) = P_M$, and*

$$\frac{\rho_k(M) - P_M}{P_M} = \Sigma_k(M) + R_k(M), \tag{2}$$

where $R_k(M) \ge 0$ satisfies

$$R_k(M) \le \frac{1}{2}\,\Sigma_{k/2}(M)^2\, e^{\Lambda(M)}. \tag{3}$$

In particular $R_k(M) \to 0$ and $\Sigma_k(M) \to 0$ as $k \to \infty$, establishing $\rho_k(M) \to P_M$.

Remark 3. The bound (3) establishes $R_k(M) \to 0$ at exponential rate at fixed M, but as an asymptotic equivalence it gives only $R_k(M) = O(\Sigma_{k/2}(M)^2)$, which at fixed M is $O(1) \cdot \Sigma_k(M)$ rather than $o(1) \cdot \Sigma_k(M)$. A tempting independence bound, $\beta_T(M) \leq \prod_{p \in T} q_p(M)$, would give the sharper conclusion but fails on the finite sampling space. For example, at $(M, T) = (17, \{3, 5\})$, direct enumeration gives $\beta_T = 9/16$ while $q_3 q_5 = (11/16)(13/16) = 143/256$; the events $\{p \mid X\}$ are not independent. The worst violation observed over $M \leq 50$ is $1/90$ at $(M, T) = (31, \{2, 3, 7\})$. The proof below instead combines the elementary geometric-mean estimate of Lemma 10 with a decomposition by $|T|$ and a uniform summation argument.

Theorem 4 (Sharp form at fixed M)**.** *For fixed $M \geq 3$, define*

$$q^*(M) := \max_{p \leq M} q_p(M), \qquad q^{**}(M) := \max_{\substack{T \subseteq \mathcal{P}_M \\ |T| \geq 2}} \beta_T(M).$$

Then $q^{**}(M) < q^*(M)$ and, as $k \to \infty$,

$$\frac{R_k(M)}{\Sigma_k(M)} = O_M\left(\left(\frac{q^{**}(M)}{q^*(M)}\right)^k\right), \tag{4}$$

where the subscript in O_M records that the implicit constant may depend on the fixed value of M. In particular $R_k(M) = o(\Sigma_k(M))$ as $k \to \infty$, and

$$\rho_k(M) - P_M = P_M \, \Sigma_k(M) \left(1 + o(1)\right) \qquad (k \to \infty, \ M \text{ fixed}).$$

Moreover, for $M \geq 11$, Ramanujan's strengthening of Bertrand's postulate [11] guarantees at least two primes in $(M/2, M]$. Consequently, $q^{**}(M) = (M-3)/(M-1)$ and $q^*(M) = (M-2)/(M-1)$, so that $q^{**}(M)/q^*(M) = (M-3)/(M-2)$ and the rate of convergence is explicit.

Theorem 5 (Joint scaling, sharp form)**.** *For each fixed $c > 0$,*

$$\lim_{M \to \infty} \ln M \cdot \Sigma_{\lfloor cM \rfloor}(M) = A(c), \tag{5}$$

where

$$A(c) := \sum_{j \geq 1} e^{-cj} \ln\left(1 + \frac{1}{j}\right), \tag{6}$$

and moreover

$$\lim_{M \to \infty} \ln M \cdot \frac{\rho_{\lfloor cM \rfloor}(M) - P_M}{P_M} = A(c). \tag{7}$$

The leading-order assertion (5) is established via Mertens' second theorem, Rosser–Schoenfeld bounds, and dominated convergence. The sharp form (7) is obtained by combining the elementary geometric-mean bound of Lemma 10 with a decomposition of the multi-prime correction by subset cardinality and a uniform summation argument.

Proposition 6 (Asymptotic behaviour of A)**.** *The function $A\colon (0, \infty) \to (0, \infty)$ is smooth, strictly decreasing, and strictly convex. Moreover:*

(i) As $c \to 0^+$,

$$A(c) = -\ln c - \gamma - \frac{c \ln c}{2} + O(c),$$

where γ is the Euler–Mascheroni constant.

(ii) As $c \to \infty$,

$$A(c) = e^{-c} \ln 2 + e^{-2c} \ln \tfrac{3}{2} + e^{-3c} \ln \tfrac{4}{3} + O(e^{-4c}).$$

1.2 Organisation

Section 2 fixes notation and records elementary lemmas, including the geometric-mean bound Lemma 10. Sections 3, 4, 5, and 6 prove Theorems 1, 2, 4, and 5 respectively. Section 7 presents numerical evidence. Section 8 discusses the fluctuations problem and other extensions. Section 9 concludes.

2 Preliminaries

Notation is standard: φ is Euler's totient, p ranges over primes, $\mathcal{P}(n) = \{p : p \mid n\}$, and $\mathcal{P}_M = \{p : p \leq M\}$. We use ln (natural logarithm) throughout; the Mertens product is $P_M = \prod_{p \leq M}(1 - 1/p)$.

For $p \in \mathcal{P}_M$, every multiple of p in $\{1, \ldots, M\}$ also lies in $\{2, \ldots, M\}$ since $p \geq 2$, so $m_p = \lfloor M/p \rfloor$ equals the number of multiples of p in the sampling space. For $T \subseteq \mathcal{P}_M$, let $A_T(M) = \{n \in \{2, \ldots, M\} : \gcd(n, \prod_{p \in T} p) = 1\}$, so $\beta_T(M) = |A_T(M)|/(M-1)$, $\beta_\varnothing(M) = 1$, and $\beta_{\{p\}}(M) = q_p(M)$.

Lemma 7. *For any $T \subseteq \mathcal{P}_M$,*

$$\mathbb{P}\big(\forall i \leq k,\ X_i \in A_T(M)\big) = \beta_T(M)^k,$$

equivalently, $\mathbb{P}(\text{no prime in } T \text{ divides } L_k) = \beta_T(M)^k$.

Proof. Immediate from independence of $X_1, \ldots, X_k$ and the fact that $p \mid L_k$ iff $p \mid X_i$ for some i. $\square$

Lemma 8. *For any $\varnothing \neq T \subseteq \mathcal{P}_M$ and any $p_0 \in T$, $\beta_T(M) \leq q_{p_0}(M)$.*

Proof. $A_T(M) \subseteq \{n \in \{2, \ldots, M\} : p_0 \nmid n\}$. $\square$

Lemma 9 (Two-prime bound for $|T| \geq 2$)**.** *Let $T \subseteq \mathcal{P}_M$ with $|T| \geq 2$, and let p_0, p_1 be any two distinct elements of T. Then for every $k \geq 1$,*

$$\beta_T(M)^k \leq q_{p_0}(M)^{k/2} q_{p_1}(M)^{k/2}.$$

Proof. By Lemma 8 applied to each of p_0, p_1, $\beta_T(M)^{k/2} \leq q_{p_0}(M)^{k/2}$ and $\beta_T(M)^{k/2} \leq q_{p_1}(M)^{k/2}$. Multiply. $\square$

Lemma 10 (Geometric-mean bound for $|T| \geq 1$)**.** *Let $T \subseteq \mathcal{P}_M$ with $|T| = r \geq 1$. For every real $k > 0$,*

$$\beta_T(M)^k \leq \prod_{p \in T} q_p(M)^{k/r}. \tag{8}$$

Proof. By Lemma 8, $\beta_T(M) \leq q_p(M)$ for every $p \in T$. Multiplying these r inequalities gives $\beta_T(M)^r \leq \prod_{p \in T} q_p(M)$, and taking both sides to the positive real power k/r preserves the inequality since both sides lie in $(0, 1]$. Rewriting the right-hand side as $\prod_{p \in T} q_p(M)^{k/r}$ gives (8). $\square$

Remark 11 (Real-exponent extension of Σ_k). Although $\Sigma_k(M) = \sum_{p \leq M} q_p(M)^k/(p-1)$ was introduced at integer k in Theorem 2, the same formula with q_p^s replaced by $q_p(M)^s$ defines $\Sigma_s(M)$ for every real $s \geq 0$, since $q_p(M) \in [0, 1]$. All monotonicity and bounds used in the sequel extend to real exponents: $\Sigma_s(M)$ is nonincreasing in s, $\Sigma_0(M) = \Lambda(M)$, and $\Sigma_s(M) \leq \Lambda(M)$ uniformly for $s \geq 0$. The sharp-form proof of Theorem 5 applies Σ_s at $s = k/r$ with $r \geq 2$; for k an integer this is generally non-integer.

Remark 12. The estimate in Lemma 10 is elementary; its importance here is structural. It permits the multi-prime correction to be stratified by $|T| = r$, replacing the wasteful factor $e^{\Lambda(M)}$ in (3) by bounds involving $\Sigma_{k/r}(M)^r/r!$. The resulting cardinality decomposition, together with uniform control as r grows, is the main step in proving the sharp joint-scaling form (7).

3 Proof of Theorem 1

For each $S \subseteq \mathcal{P}_M$, let $E_S = \{\mathcal{P}(L_k) = \mathcal{P}_M \setminus S\}$ be the event that the set of primes dividing L_k is exactly $\mathcal{P}_M \setminus S$. On E_S,

$$\frac{\varphi(L_k)}{L_k} = \prod_{p \in \mathcal{P}_M \setminus S} \left(1 - \frac{1}{p}\right) = \frac{P_M}{\prod_{p \in S}(1 - 1/p)} = P_M \prod_{p \in S} \frac{p}{p-1}. \tag{9}$$

The events $\{E_S\}_{S \subseteq \mathcal{P}_M}$ partition the sample space (primes $> M$ never divide any X_i), so

$$\rho_k(M) = P_M \sum_{S \subseteq \mathcal{P}_M} \mathbb{P}(E_S) \prod_{p \in S} \frac{p}{p-1}. \tag{10}$$

Now $E_S = \{\text{no prime in } S \text{ divides } L_k\} \cap \{\text{every prime in } \mathcal{P}_M \setminus S \text{ divides } L_k\}$, so by Möbius inversion on the subset lattice,

$$\mathbb{P}(E_S) = \sum_{T\colon S \subseteq T \subseteq \mathcal{P}_M} (-1)^{|T|-|S|}\, \mathbb{P}(\text{no prime in } T \text{ divides } L_k) = \sum_{T \supseteq S} (-1)^{|T|-|S|}\, \beta_T(M)^k, \tag{11}$$

by Lemma 7. Substituting (11) into (10) and swapping the order of summation,

$$\rho_k(M) = P_M \sum_{T \subseteq \mathcal{P}_M} \beta_T(M)^k \sum_{S \subseteq T} (-1)^{|T|-|S|} \prod_{p \in S} \frac{p}{p-1}.$$

The inner sum factors over primes in T:

$$\sum_{S \subseteq T} (-1)^{|T|-|S|} \prod_{p \in S} \frac{p}{p-1} = \prod_{p \in T} \left(\frac{p}{p-1} - 1\right) = \prod_{p \in T} \frac{1}{p-1}.$$

4 Proof of Theorem 2

Starting from (1) and isolating $T = \emptyset$ (contribution P_M),

$$\frac{\rho_k(M) - P_M}{P_M} = \sum_{T \subseteq \mathcal{P}_M,\, T \neq \emptyset} \beta_T(M)^k \prod_{p \in T} \frac{1}{p-1}.$$

The $|T| = 1$ contributions sum to $\Sigma_k(M)$. Denote the sum of the $|T| \geq 2$ contributions by $R_k(M)$.

To bound $R_k(M)$, group each T with $|T| \geq 2$ by its two smallest primes. For each pair $\{p_0, p_1\}$ with $p_0 < p_1 \in \mathcal{P}_M$, let $\mathcal{T}(p_0, p_1) = \{T \subseteq \mathcal{P}_M : \min T = p_0,\ \text{second-min}\, T = p_1\}$. Then every T with $|T| \geq 2$ lies in exactly one such class, and $T \in \mathcal{T}(p_0, p_1)$ iff $T = \{p_0, p_1\} \cup U$ for some $U \subseteq \mathcal{P}_M \cap (p_1, \infty)$. By Lemma 9,

$$\begin{aligned}
R_k(M) &= \sum_{p_0 < p_1 \in \mathcal{P}_M} \sum_{T \in \mathcal{T}(p_0,p_1)} \beta_T(M)^k \prod_{p \in T} \frac{1}{p-1} \\
&\leq \sum_{p_0 < p_1} \frac{q_{p_0}(M)^{k/2}}{p_0 - 1} \cdot \frac{q_{p_1}(M)^{k/2}}{p_1 - 1} \sum_{U \subseteq \mathcal{P}_M \cap (p_1, \infty)} \prod_{p \in U} \frac{1}{p-1} \\
&\leq \sum_{p_0 < p_1} \frac{q_{p_0}(M)^{k/2}}{p_0 - 1} \cdot \frac{q_{p_1}(M)^{k/2}}{p_1 - 1} \cdot e^{\Lambda(M)},
\end{aligned}$$

using

$$\sum_{U\subseteq\mathcal{P}_M\cap(p_1,\infty)}\prod_{p\in U}\frac{1}{p-1}=\prod_{p\in\mathcal{P}_M\cap(p_1,\infty)}\left(1+\frac{1}{p-1}\right)\le\prod_{p\le M}\left(1+\frac{1}{p-1}\right)\le e^{\Lambda(M)}.$$

The inequality $\sum_{p_0<p_1}a_{p_0}a_{p_1}\le\frac{1}{2}(\sum_p a_p)^2$ with $a_p=q_p(M)^{k/2}/(p-1)$ gives (3).

For the limit: each $q_p(M)<1$ and $\mathcal{P}_M$ is finite, so $\Sigma_k(M)\to 0$ as $k\to\infty$; and $\Sigma_{k/2}(M)\to 0$ similarly, so $R_k(M)\to 0$ by (3). Hence $(\rho_k(M)-P_M)/P_M\to 0$, i.e. $\rho_k(M)\to P_M$. □

5 Proof of Theorem 4

Recall $R_k(M)=\sum_{T\subseteq\mathcal{P}_M,|T|\ge 2}\beta_T(M)^k\prod_{p\in T}\frac{1}{p-1}$.

Step 1: $q^{**}(M)<q^*(M)$ for $M\ge 3$. Fix any $T\subseteq\mathcal{P}_M$ with $|T|\ge 2$ and pick distinct $p_0,p_1\in T$. By Lemma 8, $\beta_T(M)\le q_{p_0}(M)\le q^*(M)$. We show the first inequality is strict, which gives $\beta_T<q^*$.

Strict inclusion $A_T(M)\subsetneq A_{\{p_0\}}(M)$ holds because p_1 itself, viewed as an integer in $\{2,\dots,M\}$, belongs to $A_{\{p_0\}}(M)\setminus A_T(M)$: indeed p_1 is coprime to p_0 (distinct primes) so $p_1\in A_{\{p_0\}}(M)$, but p_1 is divisible by $p_1\in T$ so $p_1\notin A_T(M)$. Hence $|A_T|<|A_{\{p_0\}}|$, i.e. $\beta_T<q_{p_0}$. Since this holds for every T with $|T|\ge 2$, $q^{**}(M)<q^*(M)$.

Step 2: bound on R_k. For every T with $|T|\ge 2$, $\beta_T\le q^{**}$, so

$$R_k(M)=\sum_{|T|\ge 2}\beta_T^k\prod_{p\in T}\frac{1}{p-1}\le(q^{**})^k\sum_{|T|\ge 2}\prod_{p\in T}\frac{1}{p-1}.$$

The last sum is at most $\prod_{p\le M}(1+1/(p-1))=\prod_{p\le M}p/(p-1)=1/P_M\le e^{\Lambda(M)}$ (Mertens). Hence

$$R_k(M)\le(q^{**}(M))^k\cdot e^{\Lambda(M)}.\tag{12}$$

Step 3: lower bound on Σ_k. Let p^* be any prime with $q_{p^*}(M)=q^*(M)$. Then

$$\Sigma_k(M)\ge\frac{(q^*(M))^k}{p^*-1}\ge\frac{(q^*(M))^k}{M-1}.\tag{13}$$

Step 4: combining. Dividing (12) by (13),

$$\frac{R_k(M)}{\Sigma_k(M)}\le(M-1)\,e^{\Lambda(M)}\left(\frac{q^{**}(M)}{q^*(M)}\right)^k.$$

Since $q^{**}<q^*$ (Step 1), the right-hand side tends to 0 geometrically in k, establishing (4).

Step 5: explicit rate for $M\ge 11$. For $M\ge 11$, by Ramanujan's strengthening of Bertrand's postulate [11] there are at least two primes in $(M/2,M]$; call them p_1,p_2. Both have $m_p=1$, so $q_{p_i}=(M-2)/(M-1)=q^*$. For $T=\{p_1,p_2\}$, $p_1p_2>(M/2)^2\ge M$ (for $M\ge 4$), so $\lfloor M/(p_1p_2)\rfloor=0$ and, by direct inclusion–exclusion, $|A_T(M)|=(M-1)-1-1=M-3$, so $\beta_T=(M-3)/(M-1)$. To see this is the maximum: any other T with $|T|\ge 2$ satisfies either (i) $T\subseteq\mathcal{P}_M\cap(M/2,M]$, in which case $\beta_T=(M-|T|-1)/(M-1)\le(M-3)/(M-1)$ since $|T|\ge 2$; or (ii) T contains some p with $m_p\ge 2$, in which case $\beta_T\le q_p\le(M-3)/(M-1)$ (since $q_p=1-m_p/(M-1)\le(M-3)/(M-1)$). Hence $q^{**}(M)=(M-3)/(M-1)$. □

6 Proof of Theorem 5

We prove (5) first and then derive (7) from Lemma 10.

6.1 Proof of (5): leading-order convergence of Σ

Partition primes by m_p-value. A prime p satisfies $m_p = j$ iff $p \in (M/(j+1), M/j]$. For such p, $q_p(M) = 1 - j/(M-1)$. Write

$$S_j(M) := \sum_{p \in \mathcal{P}_M :\, m_p = j} \frac{1}{p-1},$$

so $\Sigma_{\lfloor cM \rfloor}(M) = \sum_{j \ge 1} (1 - j/(M-1))^{\lfloor cM \rfloor} S_j(M)$.

Pointwise limit. For each fixed $j \ge 1$, using $\ln(1-x) = -x + O(x^2)$ as $x \to 0$,

$$\big(1 - j/(M-1)\big)^{\lfloor cM \rfloor} = \exp\big(\lfloor cM \rfloor (-j/(M-1) + O(1/M^2))\big) \longrightarrow e^{-cj}.$$

By the prime number theorem with a classical de la Vallée Poussin error term [14, Ch. I.3], partial summation yields the sharpened form of Mertens' second theorem

$$\sum_{p \le x} \frac{1}{p} = \ln\ln x + \beta_1 + E(x), \qquad E(x) = O\big(\exp(-\alpha\sqrt{\ln x})\big),$$

uniformly in $x \ge 2$, where $\beta_1 \approx 0.2615$ is the Meissel–Mertens constant and $\alpha > 0$ is an absolute constant (unrelated to the regime parameter c); in particular $E(x) = o(1/\ln x)$. Applying this at $x = M/j$ and $x = M/(j+1)$ (which requires $M/(j+1) \ge 2$, i.e. $j \le M/2 - 1$),

$$\sum_{p \in (M/(j+1), M/j]} \frac{1}{p} = \ln \frac{\ln(M/j)}{\ln(M/(j+1))} + E(M/j) - E(M/(j+1)). \tag{14}$$

For each fixed $j \ge 1$, the error term is $o(1/\ln M)$ as $M \to \infty$. A first-order expansion gives

$$\ln \frac{\ln M - \ln j}{\ln M - \ln(j+1)} = \frac{\ln(1+1/j)}{\ln M}\big(1 + O_j(1/\ln M)\big).$$

The discrepancy between $\sum 1/(p-1)$ and $\sum 1/p$ over the same interval equals $\sum 1/(p(p-1))$. For each fixed $j \ge 1$ this discrepancy is bounded above by $\sum_{p > M/(j+1)} 1/(p(p-1)) = O\big((j+1)/M\big)$ (since $\sum_{n > N} 1/(n(n-1)) = 1/N$ over integers), hence of order $O(1/M) = o(1/\ln M)$. Multiplying (14) by $\ln M$ and taking $M \to \infty$ at fixed j,

$$\ln M \cdot S_j(M) \longrightarrow \ln(1+1/j) \qquad (M \to \infty) \tag{15}$$

for each fixed j.

Uniform tail bound.

Lemma 13 (Uniform envelope). *For all $M \ge 289$ and all $1 \le j \le M/17$,*

$$S_j(M) \le \frac{5.04}{\ln(M/j)}.$$

In particular, for $1 \le j \le M^{1/2}$ and $M \ge 289$, $\ln M \cdot S_j(M) \le 10.08$ uniformly in j.

Proof. By the Rosser–Schoenfeld explicit bounds [12], for $x \geq 17$, $\pi(x) \leq 1.26\, x/\ln x$; the hypothesis $j \leq M/17$ ensures $M/j \geq 17$. Hence the number of primes in $(M/(j+1), M/j]$ is at most $\pi(M/j) \leq 1.26(M/j)/\ln(M/j)$. Each such prime satisfies $p > M/(j+1) \geq M/(2j)$ (using $j \geq 1$), and for $M \geq 4j$ this gives $p-1 \geq M/(4j)$, so $1/(p-1) \leq 4j/M$ uniformly over the interval (the hypothesis $j \leq M/17$ implies $M \geq 17j \geq 4j$). Therefore

$$S_j(M) \leq \pi(M/j) \cdot \frac{4j}{M} \leq 1.26 \cdot \frac{M/j}{\ln(M/j)} \cdot \frac{4j}{M} = \frac{5.04}{\ln(M/j)}.$$

For $j \leq M^{1/2}$, $\ln(M/j) \geq \frac{1}{2}\ln M$, giving $\ln M \cdot S_j(M) \leq 10.08$. $\square$

Dominated convergence. Combining the pointwise limit (15) with the envelope Lemma 13 and the inequality $(1 - j/(M-1))^{\lfloor cM \rfloor} \leq 2e^{-cj}$ (valid uniformly for $1 \leq j \leq M^{1/2}$ and $M \geq M_1(c)$), each term in the main range is bounded by $2 \cdot 10.08\, e^{-cj}/\ln M$, and multiplied by $\ln M$ the envelope $20.16\, e^{-cj}$ is summable in j. The tail $j > M^{1/2}$ contributes at most

$$\sum_{j > M^{1/2}} \bigl(1 - j/(M-1)\bigr)^{\lfloor cM \rfloor} S_j(M) \leq \Lambda(M)\, \bigl(1 - M^{-1/2}\bigr)^{\lfloor cM \rfloor} = O\bigl(\ln\ln M \cdot e^{-c\sqrt{M}/2}\bigr),$$

which is super-polynomially small (and $\Lambda(M) = \sum_p 1/(p-1) = O(\ln\ln M)$ by Mertens). By dominated convergence,

$$\ln M \cdot \Sigma_{\lfloor cM \rfloor}(M) \longrightarrow \sum_{j \geq 1} e^{-cj} \ln(1 + 1/j) = A(c). \quad \square$$

6.2 Proof of (7): the sharp form

Because of (5), it suffices to show $\ln M \cdot R_k(M) \to 0$ in the joint regime:

$$\lim_{M \to \infty} \ln M \cdot R_k(M) = 0, \qquad k = \lfloor cM \rfloor. \tag{16}$$

Size decomposition. Group $R_k(M)$ by $|T|$:

$$R_k(M) = \sum_{r \geq 2} R_k^{(r)}(M), \qquad R_k^{(r)}(M) := \sum_{T \subseteq \mathcal{P}_M,\, |T| = r} \beta_T^k \prod_{p \in T} \frac{1}{p-1}.$$

Bound for each fixed $r \geq 2$. By Lemma 10, $\beta_T^k \leq \prod_{p \in T} q_p^{k/r}$ for $|T| = r$. Hence

$$R_k^{(r)}(M) \leq \sum_{T :\, |T| = r} \prod_{p \in T} \frac{q_p^{k/r}}{p-1} = e_r\left(\left\{\frac{q_p^{k/r}}{p-1}\right\}_{p \in \mathcal{P}_M}\right) \leq \frac{\Sigma_{k/r}(M)^r}{r!}, \tag{17}$$

where e_r denotes the r-th elementary symmetric polynomial and we used the standard inequality $e_r(x_1, \ldots, x_n) \leq (x_1 + \cdots + x_n)^r / r!$ for nonnegative x_i.

The strip argument in Section 6.1 applies, without change, to any real exponent sequence $s_M \geq 0$ satisfying $s_M/M \to d > 0$; the pointwise factor then converges to e^{-dj} and the same dominating envelope applies. Taking $s_M = \lfloor cM \rfloor / r$, for which $s_M/M \to c/r$, gives

$$\ln M \cdot \Sigma_{k/r}(M) \longrightarrow A(c/r) \qquad (M \to \infty), \tag{18}$$

for each fixed $r \geq 2$; see also Remark 11.

Combining (17) and (18), for each fixed $r \geq 2$,

$$\ln M \cdot R_k^{(r)}(M) \leq \frac{(\ln M \cdot \Sigma_{k/r}(M))^r}{r!\,(\ln M)^{r-1}} \longrightarrow 0, \tag{19}$$

because the numerator converges to $A(c/r)^r$ while $(\ln M)^{r-1} \to \infty$ for every fixed $r \geq 2$.

Uniform-in-r summability. Pointwise vanishing (19) for each fixed r is not sufficient on its own to conclude $\ln M \cdot R_k(M) \to 0$, because $R_k(M) = \sum_{r\geq 2} R_k^{(r)}(M)$ is an infinite sum. We now establish a uniform-in-r estimate sufficient for term-by-term summation.

Set $v := (k/r)/(M-1) = \lfloor cM \rfloor/(r(M-1))$, so for M sufficiently large (depending on c) and $r \geq 1$, $v \geq c/(2r)$. Using $q_j^{k/r} = (1 - j/(M-1))^{k/r} \leq e^{-jv}$ and splitting the strip sum at $j = \lfloor M^{1/2} \rfloor$,

$$\Sigma_{k/r}(M) = \sum_{j\geq 1} S_j(M)\, q_j^{k/r} \leq \sum_{j=1}^{\lfloor M^{1/2} \rfloor} S_j(M)\, e^{-jv} + \Lambda(M)\big(1 - M^{-1/2}\big)^{k/r}.$$

For $M \geq 289$ and $1 \leq j \leq M^{1/2}$, Lemma 13 gives $S_j(M) \leq 10.08/\ln M$ uniformly in j; hence

$$\sum_{j=1}^{\lfloor M^{1/2} \rfloor} S_j(M)\, e^{-jv} \leq \frac{10.08}{\ln M} \cdot \frac{e^{-v}}{1 - e^{-v}} \leq \frac{10.08}{\ln M} \cdot \frac{1}{1 - e^{-v}}.$$

For every $v > 0$, the elementary inequality $1 - e^{-v} \geq v/(1+v)$ gives $1/(1 - e^{-v}) \leq 1 + 1/v \leq 1 + 2r/c$. Meanwhile, for $2 \leq r \leq r_{\max}(M) := M^{1/2}/\ln M$ and M large, $c\sqrt{M}/(2r) \geq (c/2)\ln M$, so $(1 - M^{-1/2})^{k/r} \leq e^{-c\sqrt{M}/(2r)} \leq M^{-c/2}$, and the second sum contributes at most $\Lambda(M)\, M^{-c/2} = o(1/(\ln M)^2)$. Combining, since $1 + 2r/c \leq (2 + 2/c)\, r$ for $r \geq 1$,

$$\Sigma_{k/r}(M) \leq \frac{C_u\, r}{\ln M}, \qquad 2 \leq r \leq r_{\max}(M), \tag{20}$$

for some constant $C_u = C_u(c)$ (explicitly, $C_u \leq 10.08(2 + 2/c)$) and $M \geq M_1(c)$.

Summation over r. Combining (17) and (20), for $2 \leq r \leq r_{\max}(M)$,

$$R_k^{(r)}(M) \leq \frac{\Sigma_{k/r}(M)^r}{r!} \leq \frac{(C_u r/\ln M)^r}{r!} \leq \left(\frac{C_u e}{\ln M}\right)^r$$

where we used Stirling's bound $r! \geq (r/e)^r$. For $r > r_{\max}(M)$, the universal bound $\Sigma_s(M) \leq \Sigma_0(M) = \Lambda(M)$ (valid for any $s \geq 0$ since $q_p \in [0,1]$) gives

$$R_k^{(r)}(M) \leq \frac{\Lambda(M)^r}{r!} \leq \left(\frac{e\,\Lambda(M)}{r}\right)^r \leq \left(\frac{e\,\Lambda(M)\ln M}{M^{1/2}}\right)^r,$$

which is super-polynomially small in M (since $\Lambda(M)\ln M/M^{1/2} \to 0$) and sums to $o(1/M^{1/3})$ over $r > r_{\max}(M)$. Therefore, for M sufficiently large that $C_u e/\ln M \leq 1/2$,

$$R_k(M) \leq \sum_{r=2}^{\lfloor r_{\max} \rfloor} \left(\frac{C_u e}{\ln M}\right)^r + o\big(1/M^{1/3}\big) \leq \frac{2(C_u e)^2}{(\ln M)^2} + o\big(1/M^{1/3}\big) = O\bigg(\frac{1}{(\ln M)^2}\bigg). \tag{21}$$

Conclusion. From (21), $\ln M \cdot R_k(M) = O(1/\ln M) \to 0$ as $M \to \infty$, which is (16).

Combining with (5),

$$\ln M \cdot \frac{\rho_k - P_M}{P_M} = \ln M \cdot \Sigma_k + \ln M \cdot R_k \longrightarrow A(c) + 0 = A(c),$$

which is (7). □

Proof of Proposition 6. For every $c_0 > 0$, the series (6) and all of its termwise derivatives converge uniformly on $[c_0, \infty)$ by the Weierstrass M-test. Thus A is smooth on $(0, \infty)$, and its first two derivatives are

$$A'(c) = -\sum_{j\geq 1} j\, e^{-cj} \ln(1+1/j), \quad A''(c) = \sum_{j\geq 1} j^2\, e^{-cj} \ln(1+1/j),$$

both with every term positive, so $A'(c) < 0$ (strict monotonicity) and $A''(c) > 0$ (strict convexity) for all $c > 0$.

Proof of (ii). Isolating the first three terms of (6) and using $\ln(1+1/j)$ is decreasing in j,

$$A(c) - e^{-c}\ln 2 - e^{-2c}\ln\tfrac{3}{2} - e^{-3c}\ln\tfrac{4}{3} = \sum_{j\geq 4} e^{-cj}\ln(1+1/j) \leq \ln\tfrac{5}{4}\sum_{j\geq 4} e^{-cj} = \frac{\ln(5/4)}{1-e^{-c}}\, e^{-4c},$$

which is $O(e^{-4c})$ as $c \to \infty$ since $1/(1-e^{-c}) \to 1$.

Proof of (i). For every $j \geq 1$, the alternating Taylor series $\ln(1+x) = \sum_{k\geq 1}(-1)^{k+1}x^k/k$ at $x = 1/j$ gives

$$\ln(1+1/j) = \frac{1}{j} - \frac{1}{2j^2} + R(j), \qquad |R(j)| \leq \frac{1}{3j^3}, \tag{22}$$

where the bound on $R(j)$ is the alternating-series remainder. Substituting into (6) and separating the three resulting convergent series,

$$A(c) = -\ln(1-e^{-c}) - \tfrac{1}{2}\,\mathrm{Li}_2(e^{-c}) + \sum_{j\geq 1} e^{-cj}R(j), \tag{23}$$

where $\sum_{j\geq 1} e^{-cj}/j = -\ln(1-e^{-c})$ and $\sum_{j\geq 1} e^{-cj}/j^2 = \mathrm{Li}_2(e^{-c})$.

The sum $\sum_{j\geq 1} R(j)$ converges absolutely by (22). Since $\sum_{j=1}^{N}\ln(1+1/j) = \ln(N+1)$ (telescoping), $\sum_{j=1}^{N} 1/j = H_N = \ln N + \gamma + O(1/N)$, and $\sum_{j=1}^{N} 1/(2j^2) = \pi^2/12 + O(1/N)$,

$$\sum_{j=1}^{N} R(j) = \sum_{j=1}^{N}\big(\ln(1+1/j) - 1/j + 1/(2j^2)\big) = \ln(N+1) - H_N + \tfrac{1}{2}\sum_{j=1}^{N} 1/j^2,$$

which tends to $\pi^2/12 - \gamma$ as $N \to \infty$. Hence

$$\sum_{j\geq 1} R(j) = \frac{\pi^2}{12} - \gamma. \tag{24}$$

For the c-dependence of the third term in (23),

$$\sum_{j\geq 1} e^{-cj}R(j) - \sum_{j\geq 1} R(j) = -\sum_{j\geq 1}(1-e^{-cj})R(j),$$

whose absolute value is bounded, using $1 - e^{-cj} \leq cj$ and $|R(j)| \leq 1/(3j^3)$, by $(c/3)\sum_{j\geq 1} 1/j^2 = c\pi^2/18$. Hence

$$\sum_{j\geq 1} e^{-cj}R(j) = \frac{\pi^2}{12} - \gamma + O(c). \tag{25}$$

Finally, expand the first two terms of (23). From $1 - e^{-c} = c(1 - c/2 + O(c^2))$,

$$-\ln(1-e^{-c}) = -\ln c - \ln(1 - c/2 + O(c^2)) = -\ln c + c/2 + O(c^2). \tag{26}$$

For the dilogarithm, the reflection identity $\mathrm{Li}_2(z) + \mathrm{Li}_2(1-z) = \pi^2/6 - \ln z \ln(1-z)$ gives, at $z = e^{-c}$,

$$\mathrm{Li}_2(e^{-c}) = \frac{\pi^2}{6} + c\ln(1-e^{-c}) - \mathrm{Li}_2(1-e^{-c}).$$

Using $\ln(1-e^{-c}) = \ln c - c/2 + O(c^2)$ (from (26)) and $\mathrm{Li}_2(u) = u + u^2/4 + O(u^3)$ as $u \to 0$ with $u = 1 - e^{-c} = c - c^2/2 + O(c^3)$,

$$\mathrm{Li}_2(e^{-c}) = \frac{\pi^2}{6} + c\ln c - c - \frac{c^2}{4} + O(c^3). \tag{27}$$

Substituting (26), (27), and (25) into (23),

$$A(c) = \left[-\ln c + c/2\right] - \tfrac{1}{2}\left[\tfrac{\pi^2}{6} + c\ln c - c\right] + \left[\tfrac{\pi^2}{12} - \gamma\right] + O(c) = -\ln c - \gamma - \tfrac{c\ln c}{2} + O(c),$$

where the c-linear and c^2 remainders are absorbed into $O(c)$. □

7 Numerical Evidence

The theorems in Sections 3–6 are proved rigorously and require no numerical verification. The numerics below are included to illustrate the finite-M behaviour of the asymptotics in Theorems 2–5, which may be useful to readers who wish to work with these results computationally. In particular, the $O(1/\ln M)$ convergence in Theorem 5 is slow, and the tables below make the rate of convergence concrete. Computations and code supporting all figures and tables are openly available at the repository cited in Section 9.

Computational protocol. Exact finite-M checks use integer or rational arithmetic, and prime sums are evaluated directly. Monte Carlo estimates use NumPy's default random-number generator with $N = 2000$ independent replicates per displayed cell and deterministic cell seed $2026 + 1000M + k$. The statistic $\varphi(L_k)/L_k$ is evaluated from the union of the prime divisors of the sampled integers. Exact values obtained from Theorem 1 are computed by direct enumeration over subsets $T \subseteq \mathcal{P}_M$. The source bundle contains the scripts, cell-level standard errors, seeds, and raw CSV output.

Theorem 1. For $M \in \{5,6,7,8,9,10\}$ and $k \in \{2,3,4,5\}$, $\rho_k(M)$ computed via direct enumeration of all $(M-1)^k$ tuples was compared with the right-hand side of (1), using exact rational arithmetic throughout. All 24 test cases matched *exactly* as rationals. Against Monte Carlo (10^4 samples, standard error $\approx 10^{-3}$), agreement was within 2 standard errors.

Theorem 2. Figure 1 shows $\rho_k(M)$ versus k for $M \in \{100, 300, 1000\}$; Table 1 compares the observed gap $\rho_k(M) - P_M$ (Monte Carlo, $N = 2000$) with $P_M \Sigma_k(M)$ (computed exactly), and reports the standard error of every Monte Carlo estimate.

Theorem 4 (sharp form at fixed M). Table 2 exhibits $R_k(M)/\Sigma_k(M)$ for $M \in \{20, 30\}$ and $k \in \{20, 40, 80, 160\}$ alongside the theoretical upper bound $(q^{**}/q^*)^k$ from Theorem 4 (ignoring the $(M-1)e^{\Lambda(M)}$ prefactor). The observed ratio decays exponentially, in fact substantially faster than the bound requires: at $M = 20$, the ratio falls from 4.5×10^{-2} at $k = 20$ to 1.2×10^{-5} at $k = 160$, while the bound predicts decay by factor $(17/18)^{140} \approx 3 \times 10^{-4}$.

Theorem 5. Figure 2 plots $\ln M \cdot \Sigma_{\lfloor cM \rfloor}(M)$ (computed exactly) at $M \in \{10^3, 10^4, 10^5, 10^6\}$ against $A(c)$; Table 3 extends to $M = 10^7$ across 15 values of $c \in [0.05, 15]$. The relative error $(\ln M \cdot \Sigma_{\lfloor cM \rfloor}(M) - A(c))/A(c)$ at $M = 10^7$ ranges from approximately 10.7% at $c = 0.05$ down to 2.1% at $c = 15$, decreasing monotonically in c; these residuals are consistent with an $O(1/\ln M)$ correction arising from the next-order expansion of the prime-strip difference in (14).

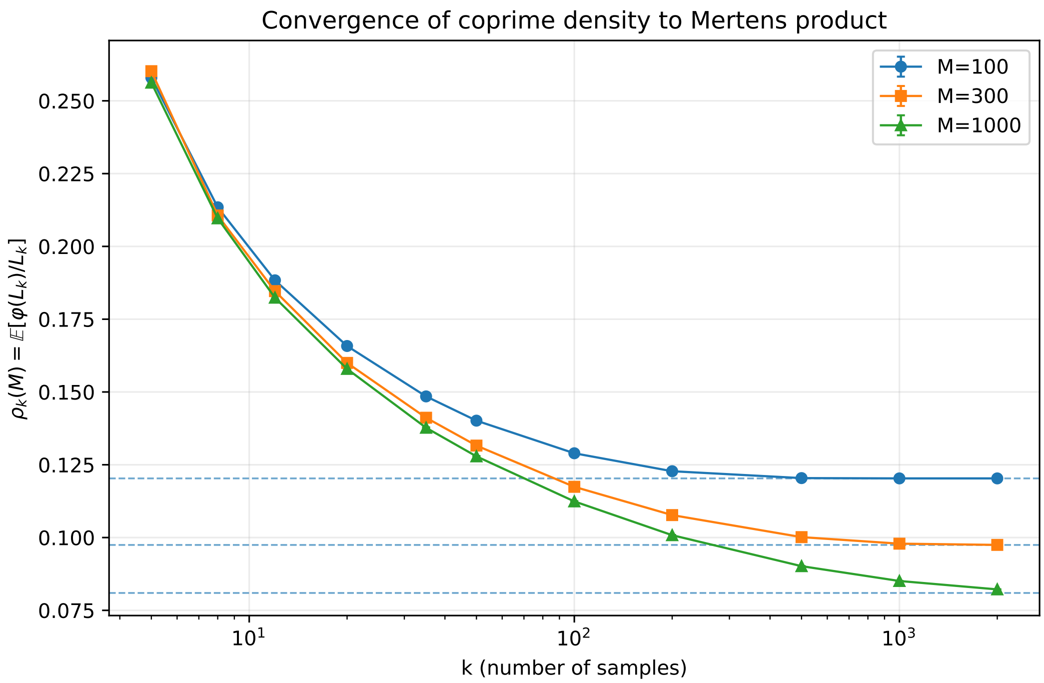

Figure 1: Monte Carlo estimates of $\rho_k(M)$ for $M \in \{100, 300, 1000\}$, based on $N = 2000$ independent replicates per point. Error bars show ± 1 standard error of the sample mean. Dashed lines indicate the exact Mertens product P_M.

Table 1: Deterministic Monte Carlo estimates of $\rho_k(M) - P_M$ ($N = 2000$ per cell) compared with the exact first-order prediction $P_M \Sigma_k(M)$. The reported standard errors are cell-specific. The ratio column illustrates the approach to unity predicted by Theorem 2.

M	k	$\rho_k(M)$ (MC)	s.e.	P_M	gap obs.	gap pred.	ratio
100	50	0.140151	1.41×10^{-4}	0.120317	0.019834	0.018552	1.069
100	200	0.122783	4.75×10^{-5}	0.120317	0.002465	0.002482	0.993
300	100	0.117427	7.59×10^{-5}	0.097459	0.019969	0.018264	1.093
300	500	0.100147	2.43×10^{-5}	0.097459	0.002688	0.002642	1.018
1000	500	0.090173	2.32×10^{-5}	0.080965	0.009208	0.008758	1.051
1000	2000	0.082197	8.33×10^{-6}	0.080965	0.001232	0.001220	1.010
1000	5000	0.081022	1.82×10^{-6}	0.080965	0.000057	0.000055	1.032
3000	5000	0.071308	4.63×10^{-6}	0.069966	0.001342	0.001335	1.006

Table 2: Observed ratio $R_k(M)/\Sigma_k(M)$ versus theoretical rate $(q^{**}(M)/q^*(M))^k$ from Theorem 4. The bound is not sharp but captures the exponential decay. For $M \geq 11$, $q^{**}/q^* = (M-3)/(M-2)$.

M	k	R_k/Σ_k (obs.)	$(q^{**}/q^*)^k$ (rate)
20	20	4.52×10^{-2}	3.19×10^{-1}
20	40	1.22×10^{-2}	1.02×10^{-1}
20	80	1.15×10^{-3}	1.03×10^{-2}
20	160	1.18×10^{-5}	1.07×10^{-4}
30	20	6.30×10^{-2}	4.83×10^{-1}
30	40	2.27×10^{-2}	2.34×10^{-1}
30	80	4.30×10^{-3}	5.45×10^{-2}
30	160	2.20×10^{-4}	2.97×10^{-3}

Linear extrapolation of the sequence $\{\ln M \cdot \Sigma_{\lfloor cM \rfloor}(M)\}_M$ in the variable $1/\ln M$ (fitted over $M \in \{10^4, 10^5, 10^6, 10^7\}$) agrees with $A(c)$ to within approximately 0.5% for $c \in [0.1, 5]$ and to within 1.5% across the full tested range $c \in [0.05, 15]$.

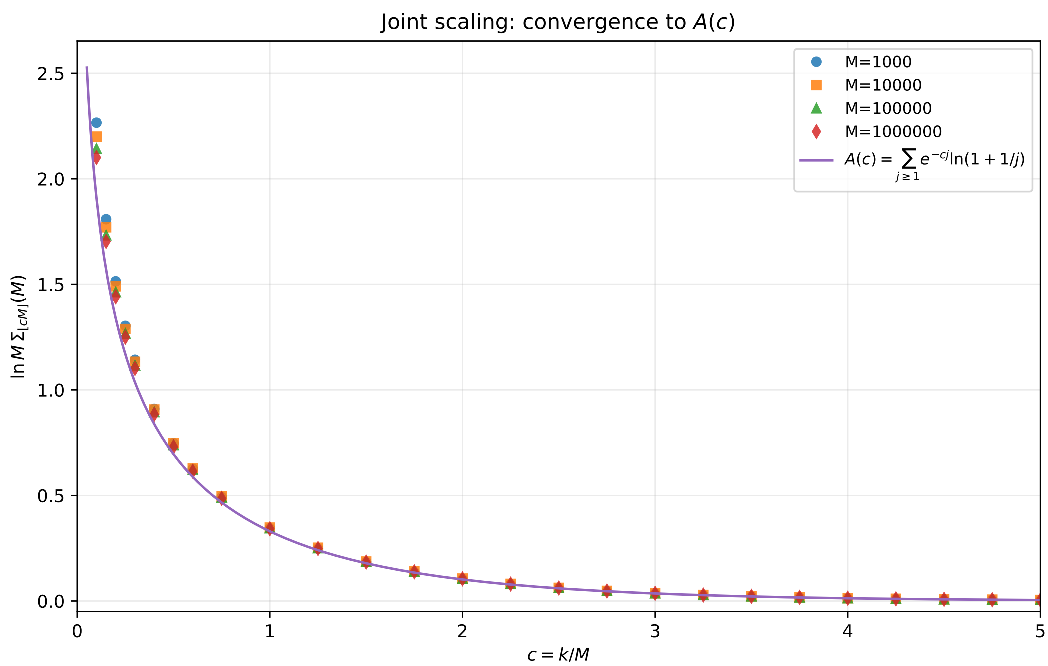

Figure 2: Joint scaling regime $k = \lfloor cM \rfloor$. Points show $\ln M \cdot \Sigma_{\lfloor cM \rfloor}(M)$, computed exactly (no Monte Carlo) at $M \in \{10^3, 10^4, 10^5, 10^6\}$. Solid curve: $A(c) = \sum_{j \geq 1} e^{-cj} \ln(1 + 1/j)$, computed to $j_{\max} = 200$.

Table 3: $\ln M \cdot \Sigma_{\lfloor cM \rfloor}(M)$ versus $A(c)$ at M up to 10^7. All values computed exactly; no Monte Carlo. Residuals decay consistently with an $O(1/\ln M)$ next-order strip correction; see text.

c	$A(c)$	$M = 10^4$	$M = 10^5$	$M = 10^6$	$M = 10^7$
0.05	2.5267	3.0277	2.9231	2.8498	2.7982
0.10	1.9095	2.1999	2.1428	2.1009	2.0714
0.15	1.5688	1.7704	1.7324	1.7033	1.6831
0.25	1.1687	1.2886	1.2671	1.2498	1.2380
0.50	0.6954	0.7470	0.7382	0.7307	0.7258
0.75	0.4665	0.4950	0.4903	0.4861	0.4835
1.00	0.3301	0.3476	0.3448	0.3422	0.3406
1.25	0.2406	0.2522	0.2504	0.2486	0.2476
1.50	0.1787	0.1867	0.1855	0.1843	0.1836
2.00	0.1020	0.1061	0.1056	0.1049	0.1045
3.00	0.0356	0.0368	0.0367	0.0365	0.0364
5.00	0.00469	0.00485	0.00483	0.00480	0.00479
7.50	0.000384	0.000396	0.000395	0.000393	0.000392
10.00	0.0000315	0.0000325	0.0000324	0.0000322	0.0000321
15.00	0.00000021	0.00000022	0.00000022	0.00000021	0.00000021

8 Discussion and Open Problems

Uniform vs. non-uniform rates. Theorem 4 gives an exponential rate at fixed M. For $M \geq 11$, the controlling ratio is

$$\frac{q^{**}(M)}{q^{*}(M)} = \frac{M-3}{M-2} \longrightarrow 1.$$

Thus the fixed-M bound degenerates as $M \to \infty$ and does not yield the sharp joint-scaling result of Theorem 5. The joint regime instead requires a uniform argument that combines Lemma 10, decomposition by subset cardinality, and summation over growing r.

Second-order joint asymptotics. The residuals in Table 3 suggest studying an expansion of the form

$$\ln M \, \frac{\rho_{\lfloor cM \rfloor}(M) - P_M}{P_M} = A(c) + \frac{B(c)}{\ln M} + o\left(\frac{1}{\ln M}\right).$$

At this scale, both the next-order expansion of the prime-strip sum and the $|T| = 2$ component of $R_k(M)$ may contribute. Determining their combined limit and proving that the terms with $|T| \geq 3$ are negligible are left open.

Second-order fluctuations. A natural next question is the scale and distributional shape of the fluctuations of $\ln(\varphi(L_k)/L_k)$ about its mean in the same joint regime: the variance asymptotic and an associated central limit theorem. These are developed in separate work in preparation.

General distributions. Theorem 1 generalises verbatim to any distribution on $\{2, \ldots, M\}$, with $\beta_T(M)$ replaced by the probability of being coprime to $\prod_{p \in T} p$. Theorem 2 generalises similarly. The joint scaling (5) depends on the distribution of m_p-values, which is distribution-dependent.

Applications to non-uniform sampling diagnostics. For distributions close to uniform, the function $A(c)$ is robust, but substantial departures from uniformity deform it in a distribution-specific way. Empirical estimates of $A(c)$ could therefore serve as a statistical test of uniformity for random integer generators over bounded ranges; a systematic development is left to future work.

9 Conclusion

We have introduced the expected coprime density $\rho_k(M) = \mathbb{E}[\varphi(L_k)/L_k]$ of the LCM of random integers in the regime of fixed support and growing sample size, and proved four theorems and one proposition. Theorem 1 establishes an exact finite-sum identity over subsets of $\mathcal{P}_M$. Theorem 2 shows $\rho_k(M) \to P_M$ at fixed M with explicit rate. Theorem 4 sharpens this to a full asymptotic equivalence at fixed M, with explicit geometric rate controlled by the ratio $q^{**}(M)/q^*(M)$ of the second-largest to largest single-prime coprime densities. Theorem 5 establishes the sharp joint-scaling equivalence $\ln M \cdot (\rho_{\lfloor cM \rfloor}(M) - P_M)/P_M \to A(c) := \sum_{j \geq 1} e^{-cj} \ln(1 + 1/j)$. The uniform control of the multi-prime correction in Theorem 5 comes from combining the elementary bound $\beta_T^k \leq \prod_{p \in T} q_p^{k/|T|}$ with a decomposition by $|T|$ and a summation argument uniform in the subset size. Proposition 6 gives the $c \to 0^+$ and $c \to \infty$ asymptotics of A. Numerical computation up to $M = 10^7$ illustrates Theorem 5. The regime studied here complements earlier fixed-size tuple, random-subset, and joint-limit models through its i.i.d. sampling-with-replacement formulation and its focus on coprime density; the exact subset identity (Theorem 1) may be of independent use in related problems involving multiplicative functions of random LCMs.

Data and code availability. Source code, deterministic seeds, cell-level Monte Carlo errors, and raw numerical output for all figures and tables are available at `https://github.com/Wakid90/coprime-density-lcm`. The versioned preprint is archived at doi:10.5281/zenodo.21325880.

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